

M38223E4HP@QFP80(1).BIN analysis report

Executive identification

- Chip/dump size: Mitsubishi M38223E4HP, 3822 group 8-bit MCU, MELPS 740 family; file size 16,254 bytes / 0x3F7E.
- Likely role: Power-sense microcontroller firmware, Rev 8, RIOS Systems.
- Confidence: High for role; medium for exact vector names.
- SHA-256: 96c6e37cfa52f30b303db70c2036cbf21e6e1bb638c5eb11343ab161db3c9cc0
- MD5: f9a32ab6985b5cc71100b789b98fb10a
- Byte entropy: 5.3790 bits/byte.

Evidence from this dump

- Starts with the banner M3822X POWER SENSE MICON FIRMWARE Rev 8 (C) 1995 RIOS SYSTEMS CO., LTD..
- The contents decode plausibly as Mitsubishi 740-family / 65C02-like MCU code after the banner.
- If the image tail is aligned to address 0xFFFF, the inferred base is 0xC082 and the tail contains little-endian vector-like words from about 0xFFE0 to 0xFFFF.
- Several vectors point to common stubs or handlers. The exact vector names are tentative because the dump length is slightly under 16 KiB and family-specific vector layout matters.

Structural notes

- Active 4 KiB ranges (heuristic, excludes mostly 0x00/0xFF blocks): 0x00000-0x02FFF
- Longest 0xFF runs: 0x287E len 5854, 0x23A1 len 51, 0x24C8 len 5, 0x51B len 4, 0xE4E len 2
- Longest 0x00 runs: 0x2555 len 60, 0x245B len 37, 0x2481 len 32, 0x24CD len 17, 0x13D7 len 8
- PC option-ROM signatures found:

(none)

Tail vector table, tentative

- Base assumption: image tail maps to 0xFFFF, base = 0xC082.
- 0xFFE0: 0xD612 (inside image, file offset 0x3F5E)
- 0xFFE2: 0xE8FC (inside image, file offset 0x3F60)
- 0xFFE4: 0xD190 (inside image, file offset 0x3F62)
- 0xFFE6: 0xD177 (inside image, file offset 0x3F64)
- 0xFFE8: 0xE8FC (inside image, file offset 0x3F66)
- 0xFFEA: 0xE8FC (inside image, file offset 0x3F68)
- 0xFFEC: 0xE8FC (inside image, file offset 0x3F6A)
- 0xFFEE: 0xE307 (inside image, file offset 0x3F6C)
- 0xFFFF0: 0xE366 (inside image, file offset 0x3F6E)
- 0xFFFF2: 0xD8D4 (inside image, file offset 0x3F70)
- 0xFFFF4: 0xE8FC (inside image, file offset 0x3F72)
- 0xFFFF6: 0xE2AA (inside image, file offset 0x3F74)
- 0xFFFF8: 0xE2CF (inside image, file offset 0x3F76)
- 0xFFFFA: 0xDC61 NMI? (inside image, file offset 0x3F78)

- 0xFFFFC: 0xD1AF RESET? (inside image, file offset 0x3F7A)
- 0xFFFFE: 0xE6B1 IRQ/BRK? (inside image, file offset 0x3F7C)

Interesting strings

- 0x000000: M3822X POWER SENSE MICON FIRMWARE Rev 8 (C) 1995 RIOS SYSTEMS CO.,LTD.x ^

Included supplemental files

- strings/M38223E4HP_power_sense_mcu.strings.txt - all printable strings found by a simple ASCII scanner.
- hexdumps/M38223E4HP_power_sense_mcu_selected_hexdumps.txt - selected annotated hexdumps.
- disassembly/M38223E4HP_power_sense_mcu_reachable_melps740_approx_disasm.txt
- disassembly/M38223E4HP_power_sense_mcu_linear_melps740_approx_disasm.txt

Caveats

- This is static triage, not a full reverse-engineering project. Hardware register names, MCU peripheral vector names, and exact memory maps require board schematics, data books, or emulator instrumentation.
- The MELPS 740 disassembler is approximate. It decodes many 6502/65C02-like opcodes and common bit operations, but Mitsubishi-specific opcodes may be mislabeled or emitted as data bytes.